寬數週練 吳寬老師出題・範例題本　　建議作答 30 分鐘

SAT Math　綜合範例

學生題本

先讀「觀念提示」，再依序作答。寫完再請家長對答。　　Digital SAT Math

範例題本　這份是讓家長先看題目風格的範例（10 題）。**實際題本依孩子需求客製**：家長填寫孩子的年級、學校進度與弱點後，吳寬老師再出專屬題本。

姓名 _____　　日期 _____

觀念提示：作答提醒

1. 題幹多為英文，同正式 Digital SAT；[MC] 選擇題四選一，[SPR] 自己填答。
2. Read what the question asks：題目問的常常不是 x，而是某個組合或參數。
3. 後 5 題是第二模組（難）的題型，卡住是正常的，先寫下列出的式子。

範例 10 題

Easy 2・Medium 3・Hard 5（第 6–10 題為 Module 2 hard 等級）

1. ［Easy・一次方程式］ **[SPR]** If $3(x-2)+5=2x+7$, what is the value of x? _____

2. ［Easy・模型解讀］ **[MC]** A plumber's total charge, in dollars, for a job lasting h hours is $C = 45 + 30h$. What is the best interpretation of 30 in this context?

A. The total charge for a 1-hour job　**B.** The charge per hour　**C.** The fixed fee charged for every job　**D.** The number of hours in a job

Answer: _____

3. ［Medium・次數表］ **[SPR]** The table shows how many books each of 20 students read last month.

Books	0	1	2	3	4
Students	3	5	7	4	1

What is the median minus the mean? _____

4. ［Medium・相似三角形］ **[SPR]** In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. If $AD = 4$, $DB = 6$, and the area of triangle ADE is 12, what is the area of quadrilateral $DBCE$?

_____

5. ［Medium・指數模型等價式］ **[MC]** A town's population is modeled by $P(t) = 5000(1.44)^t$, where t is the number of years since 2020. Which equivalent expression shows that the population increases by 20% every 6 months?

A. $5000(1.2)^{t/2}$ **B.** $5000(1.2)^{2t}$ **C.** $5000(1.72)^t$ **D.** $5000(0.72)^{2t}$

Answer: _____

6. ［Hard・圓與切線］ **[SPR]** In the xy-plane, the graph of $x^2 + y^2 - 6x + 8y = k$ is a circle that is tangent to the x-axis. What is the value of k? _____

7. ［Hard・無交點條件］ **[MC]** In the xy-plane, the line $y = 2x + k$ and the parabola $y = x^2 - 4x + 5$ have no points of intersection. Which of the following could be the value of k?

A. -5 **B.** -4 **C.** 0 **D.** 4

Answer: _____

8. ［Hard・恆正條件］ **[SPR]** The function f is defined by $f(x) = x^2 - 2ax + 3a + 4$, where a is a constant. If $f(x) > 0$ for all real values of x, how many integer values of a are possible? _____

9. ［Hard・整數根參數］ **[SPR]** The equation $x^2 + bx + 36 = 0$, where b is a constant, has two distinct solutions, and both solutions are positive integers. What is the greatest possible value of b? _____

10. ［Hard・弦長］ **[SPR]** In the xy-plane, the line $y = 2$ intersects the circle $(x - 2)^2 + (y + 1)^2 = 25$ at two points. What is the distance between these two points? _____

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